

Project Design Phase Solution Architecture

Date	4 July 2025
Team ID	LTVIP2025TMID49900
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

1. Objectives of the Solution Architecture

- Bridge Business and Technology Needs: Enable stakeholders (researchers, policymakers, tourism boards) to access meaningful insights about UNESCO World Heritage Sites.
- Best Fit Technology: Use Tableau for visual analytics, Tableau Prep/Excel for preprocessing, and cloud hosting for accessibility.
- Clear Structure & Behavior: Define how data flows from source to dashboards, ensuring accuracy, interactivity, and performance.
- Phased Delivery: Design solution to be developed iteratively using sprints (data preparation → dashboard creation → story building → optimization).

2. Solution Architecture Overview

Layers

1. Data Layer (Input):

- UNESCO World Heritage Sites dataset (2019).
- Attributes: Country, Region, Type (Cultural/Natural/Mixed), Danger Status, Date of Inscription.

2. Processing Layer (Transformation):

- Tableau Prep / Excel: Clean and standardize data (remove nulls, format columns, correct inconsistencies).
- Calculation Fields in Tableau: Create derived metrics like total sites per country, endangered site percentage, and year-over-year growth.

3. Visualization Layer (Output):

- Tableau dashboards with:
 - Country-wise distribution (Treemap/Bar Chart)
 - Endangered vs Safe Sites (Pie Chart)
 - Regional Inscription Trends (Line Chart)
- Tableau Story combining the dashboards for guided exploration.

4. Interaction Layer:

- Filters for Region, Type (Cultural/Natural/Mixed), Danger Status, and Year of Inscription.
- Tooltips, highlights, and drill-downs for interactive exploration.

5. Hosting & Access Layer:

- Tableau Public for global accessibility.
- Embedded dashboard option for websites (future enhancement).